ANSWER KEY — Integration Lesson 6 Test: The Definite Integral, the Official Symbol

For the grader · full working shown · give credit for correct method with small arithmetic slips

1. Rain falls at a rate of r(t) millimeters per hour. Write the definite integral that means "the total rain that fell from hour 1 to hour 5."

> **Work:** start = 1 (lower limit), end = 5 (upper limit), rate = r(t), variable is t so the slice width is dt. **Answer:** $\int_1^5 r(t)\,dt$**.**

2. In the integral $\int_0^8 (3x + 2)\,dx$, identify: (a) the integrand, (b) the lower limit, (c) the upper limit, and (d) what the dx stands for.

> **Work:** the integrand is the function between the ∫ and the dx; the limits are the start/end values on the x-axis. **Answer: (a) 3x + 2; (b) 0; (c) 8; (d) dx is "a tiny little bit of x" — the infinitely skinny width of each slice.**

3. Evaluate $\int_1^5 6\,dx$ using geometry. Name the shape you used.

> **Work:** constant integrand → rectangle. Width = 5 − 1 = 4, height = 6. Area = 6 × 4 = 24. **Answer: 24 (a rectangle).**

4. Evaluate $\int_0^3 4x\,dx$ using geometry. Name the shape you used.

> **Work:** the line 4x rises from height 0 (at x = 0) to height 4 × 3 = 12 (at x = 3) → triangle with base 3, height 12. Area = ½ × 3 × 12 = 18. **Answer: 18 (a triangle).**

5. Evaluate $\int_5^5 (x^3 + 9)\,dx$, and explain in one sentence why no computing is needed.

> **Work:** the region runs from x = 5 to x = 5 — width 5 − 5 = 0, so it encloses no area regardless of the integrand. **Answer: 0, because a zero-width region has zero area (**$\int_a^a f\,dx = 0$ **always).**

6. Evaluate $\int_0^4 (x + 2)\,dx$ using geometry.

> **Work:** heights at the ends: at x = 0, height = 2; at x = 4, height = 6. Trapezoid: ½(2 + 6) × 4 = ½ × 8 × 4 = 16. (Or rectangle 2 × 4 = 8 plus triangle ½ × 4 × 4 = 8 → 16.) **Answer: 16.**

7. A car's speed after t hours is v(t) miles per hour. Translate $\int_2^6 v(t)\,dt$ into a plain-English sentence, including the units of the answer.

> **Work:** limits 2 → 6 hours; integrand is a speed, so each slice v(t) × dt is a tiny distance; the sum is total distance. Units: (miles/hour) × hours = miles. **Answer: "the total distance, in miles, the car travels between t = 2 and t = 6 hours."**

8. Given $\int_0^3 f(x)\,dx = 8$ and $\int_3^9 f(x)\,dx = 11$. Find $\int_0^9 f(x)\,dx$, and name the property.

> **Work:** the pieces 0→3 and 3→9 fit together with no gap or overlap, so $\int_0^9 f = \int_0^3 f + \int_3^9 f = 8 + 11 = 19$. **Answer: 19, by the splitting property (chop the region at x = 3).**

9. Given $\int_0^{12} g(x)\,dx = 40$ and $\int_0^5 g(x)\,dx = 15$. Find $\int_5^{12} g(x)\,dx$.

> **Work:** whole = left + right: $40 = 15 + \int_5^{12} g$, so $\int_5^{12} g = 40 - 15 = 25$. **Answer: 25.**

10. Flow rate climbs along a straight line from 0 (at x = 0) to 6 (at x = 3), then holds steady at 6 from x = 3 to x = 5. Evaluate $\int_0^5 f(x)\,dx$ with units.

> **Work:** split at x = 3. Piece 1 (ramp): triangle with base 3, height 6 → ½ × 3 × 6 = 9. Piece 2 (flat): rectangle, width 5 − 3 = 2, height 6 → 12. Total = 9 + 12 = 21. Units: (liters/minute) × minutes = liters. **Answer: 21 liters — the total water delivered in the 5 minutes.**